NOVEX HACKATHON 2026

AQUA PINCH

Automated n8n & LLM Orchestration for Industrial Water Pinch Optimization

PS CATEGORY: Water Reuse & Recycling in Chemicals & Petrochemicals

Problem Statement: Traditional water pinch analysis relies on manual source-sink matching and checking hundreds of piping permutations, creating slow, error-prone workflows that delay industrial water reuse.

TEAM & SUBMISSION DETAILS

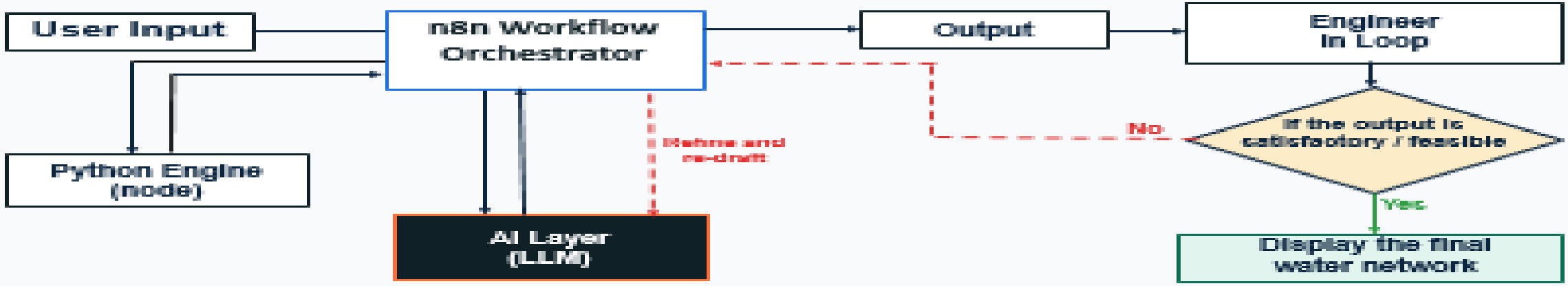
TEAM NAME
AQUA PINCH Team

PRESENTERS
Swapan Chakraborty | Uzma Lambay

INSTITUTION
Dept. of Chemical Engineering, Thadomal Shahani Engineering College, Mumbai

IDEA TITLE:- AQUA PINCH

Integrated AI-Pinch Framework



- **n8n Workflow Core: Automates seamless data passing between Python computational modules and LLM layout prompts.**
- **Python Engine: Performs mass balance calculations, generating composite curves & exact targeting metrics.**
- **LLM Layout Agent: Converts numerical targeting outputs into non-crossing physical water routing topologies.**
- **Human-in-the-Loop: Enables instant prompt-based revisions whenever plant physical constraints change.**

The Execution Gap & Bottleneck

Water scarcity and strict environmental limits strain plant throughput. Traditional Pinch Analysis calculates minimum targets but relies on manual source-sink matching across hundreds of pipe permutations—a slow, error-prone workflow.

Proposed AQUA PINCH Solution

An integrated AI framework combining Python with Large Language Models (LLMs) orchestrated via n8n. Converts theoretical targeting into practical, deployable pipe network diagrams in minutes.

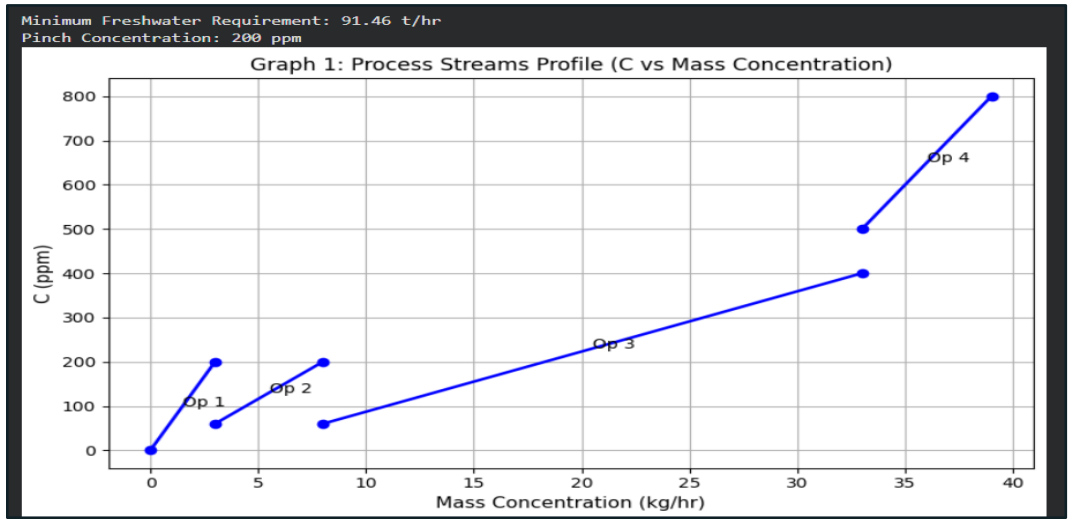
TECHNICAL APPROACH

PROOF OF CONCEPT: CASE STUDY RESULTS

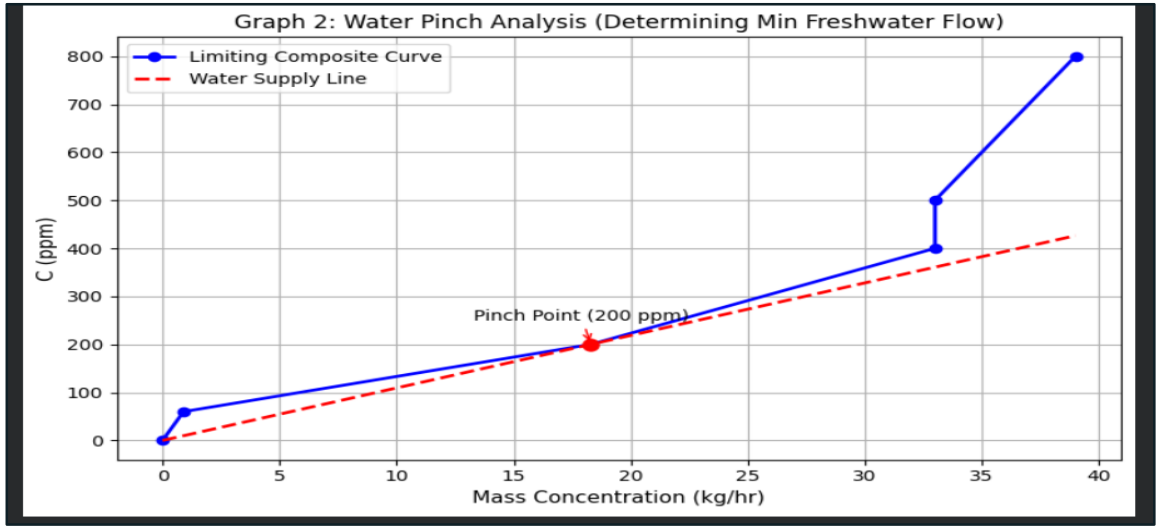
Problem Statement

Operation No.	Contaminant Mass (g/hr)	Cin (ppm)	Cout (ppm)
1.	3000	0	200
2.	5000	60	200
3.	25000	60	400
4.	6000	500	800

Results

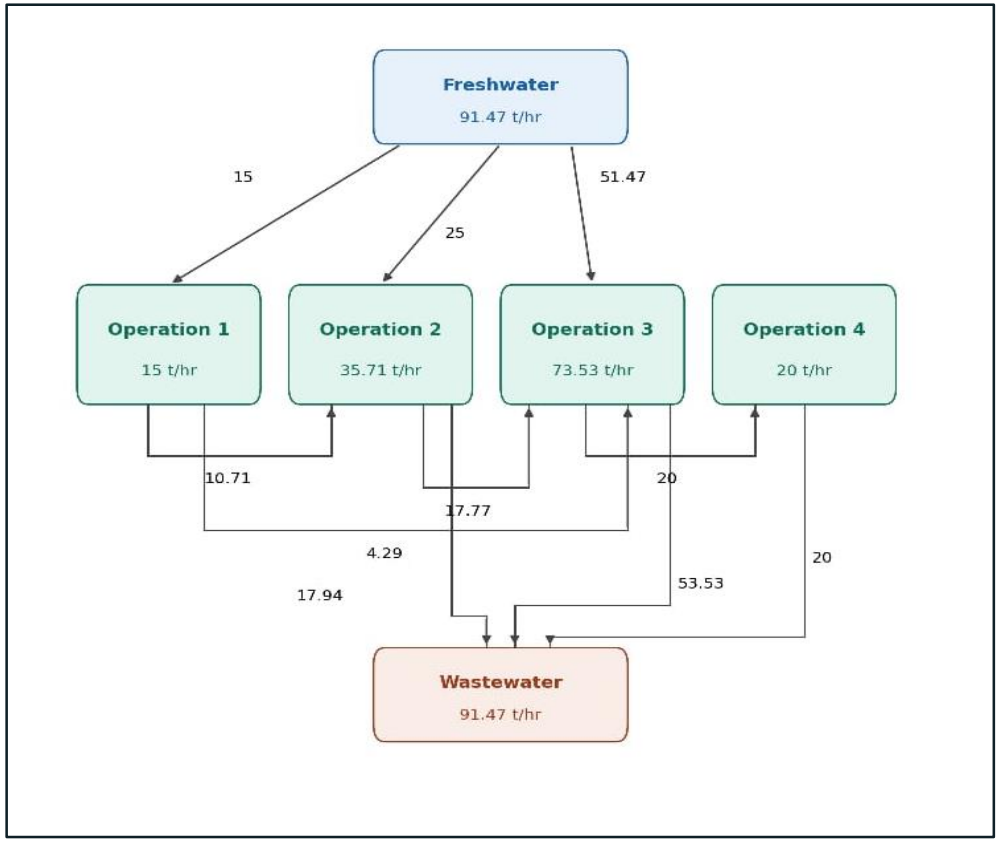


Process Streams Profile (Python)



Limiting Composite (Curve & Pinch Point)

LLM-GENERATED WATER NETWORK ARCHITECTURE



LLM-Generated Water Line Network (Claude)

Test case: We took 4 water-using processes (input of which is given in the table)

Python output: freshwater target ≈ 91.46 t/h, pinch concentration ≈200 ppm, the calculations matches our manual calculation

AI output: LLM (Claude/DeepSeek) generated an final water network .

Impact: a design that took hours manually is now completed in approximately 10–12 minutes in total

FEASIBILITY AND VIABILITY

Economic & Operational Feasibility

Zero New Hardware: Fully software-based decision tool running on existing plant IT infrastructure.

OpEx Reductions: Substantial savings on raw water procurement and effluent treatment plant (ETP) loads.

Rapid ROI: Instant payback driven by immediate reduction in plant water costs.

Risk Reduction & Accuracy

Human Error Elimination: Rigorous mass balance checks prevent stream mismatching.

Better Network Decisions: Evaluates hundreds of source-sink connections to select optimal routes.

Regulatory Compliance: Helps refineries adhere to stringent Zero Liquid Discharge (ZLD) norms.

Environmental & Social Impact

Decarbonization: Lower pumping and treatment requirements directly reduce plant energy usage and overall carbon footprint.

Conservation: Reduces industrial freshwater abstraction, preserving local groundwater tables for neighboring communities during drought periods.

IMPACT AND BENEFITS

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Target Audience Impact

Decentralized, low-cost water treatment for municipal and industrial operators.

Economic Value

Cuts utility operational costs drastically via max internal thermal energy recovery.

Environmental

Zero net external thermal waste, reducing carbon footprint & aquatic thermal pollution.

RESEARCH AND REFERENCES

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Energy Pinch Analysis
Linnhoff & Hindmarsh process heat integration for water treatment networks.

AI Cascade Optimization
Real-time stream monitoring algorithms for dynamic thermal recycling.

Modular Infrastructure
Standardized plug-and-play water treatment skid benchmarks (10k–1M GPD).

n8n & LLM Automation Framework: Open-source workflow automation & LLM orchestration models.

Industrial Water Management: Mann, J. G., & Liu, Y. A. (1999). *Industrial Water Reuse and Wastewater Minimization*.

Water Pinch Methodology: Wang, Y. P., & Smith, R. (1994). *Wastewater minimisation*. Chemical Engineering Science.